

LUBLIN RADAWIEC, Poland

WMO: 124950

Lat: 51.2167N

Lon: 22.3931E

Elev: 786

StdP: 14.28

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.6	8.3	-1.8	5.1	3.6	3.3	6.7	9.1	22.1	35.6	20.3	32.7	5.3	110	0.527

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.1	85.2	68.8	81.8	67.6	78.7	65.9	70.9	81.5	69.0	79.0	67.3	76.1	6.4	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.2	103.0	76.3	65.5	96.9	73.7	64.0	91.9	71.5	35.2	81.3	33.6	79.0	32.2	76.1	80.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
18.7	16.2	14.2	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-2.7	90.3	6.5	2.9	-7.4	92.4	-11.2	94.1	-14.9	95.7	-19.6	97.8
			-3.3	73.8	6.3	2.2	-7.8	75.3	-11.5	76.6	-15.1	77.8	-19.7	79.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	46.9	26.8	29.3	36.5	47.7	56.4	62.1	66.0	65.2	56.4	47.0	38.2	29.9	(6)
(7)		DBStd	15.84	9.86	9.50	8.37	7.77	6.90	6.09	5.80	5.80	6.49	7.59	8.07	8.92	(7)
(8)		HDD50	3019	721	578	422	131	20	0	0	0	16	150	359	622	(8)
(9)		HDD65	6831	1186	998	884	519	276	127	59	69	266	557	803	1087	(9)
(10)		CDD50	1888	0	0	3	62	219	363	497	471	209	58	6	0	(10)
(11)		CDD65	226	0	0	0	1	10	40	92	74	9	0	0	0	(11)
(12)		CDH74	2010	0	0	0	14	112	341	797	669	77	0	0	0	(12)
(13)		CDH80	488	0	0	0	2	9	61	213	194	9	0	0	0	(13)
(14)	Wind	WSAvg	6.6	7.8	7.7	8.0	7.1	5.9	5.6	5.6	5.2	5.4	6.4	7.3	7.7	(14)
(15)	Precipitation	PrecAvg	23.8	1.3	1.2	1.5	1.7	2.8	2.6	3.5	2.2	2.4	1.8	1.5	1.3	(15)
(16)		PrecMax	30.4	3.1	2.3	2.6	3.9	9.0	5.5	7.6	7.6	6.0	4.8	3.1	2.7	(16)
(17)		PrecMin	17.6	0.1	0.5	0.4	0.1	0.9	0.8	0.4	0.3	0.2	0.1	0.0	0.4	(17)
(18)		PrecStd	3.3	0.7	0.5	0.6	0.9	1.8	1.2	1.8	1.4	1.5	1.3	0.7	0.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.3	51.6	62.1	75.4	81.2	85.6	89.6	90.0	80.8	70.2	59.5	49.9	(19)
(20)		MCWB	43.8	45.8	51.5	59.2	65.1	70.8	69.9	69.7	64.8	60.4	53.6	46.9	(20)	
(21)		2%	DB	43.6	47.1	56.1	69.6	77.2	81.5	85.4	85.2	75.8	65.8	55.2	46.1	(21)
(22)		MCWB	41.2	43.4	47.7	56.1	63.7	68.4	69.0	68.4	63.2	58.0	50.8	43.9	(22)	
(23)		5%	DB	40.7	44.1	52.1	64.9	73.7	78.0	81.9	81.1	71.9	62.1	52.2	43.0	(23)
(24)		MCWB	38.7	41.3	45.5	53.4	62.1	66.1	67.9	66.9	61.2	56.4	49.1	41.2	(24)	
(25)		10%	DB	38.3	40.9	48.0	60.9	69.8	74.6	78.4	77.2	68.1	58.3	49.2	40.5	(25)
(26)		MCWB	36.7	38.4	43.1	51.0	59.4	64.0	66.4	64.8	59.4	54.0	46.9	39.0	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.2	46.7	52.2	60.0	67.9	72.4	73.5	72.7	67.0	61.9	54.3	47.8	(27)
(28)		MADB	46.5	50.0	60.6	73.2	78.0	82.6	84.3	84.5	77.2	68.1	58.3	49.6	(28)	
(29)		2%	WB	41.5	43.9	48.8	57.3	65.0	69.8	71.2	69.9	64.5	59.2	51.7	44.3	(29)
(30)		MADB	43.4	46.8	55.2	67.8	74.8	79.2	81.6	81.7	73.9	64.5	54.5	46.0	(30)	
(31)		5%	WB	39.0	41.3	45.9	54.6	62.9	67.5	69.3	68.0	62.4	56.8	49.4	41.5	(31)
(32)		MADB	40.5	43.9	50.5	63.0	72.1	75.7	79.0	78.7	69.7	61.3	51.8	42.8	(32)	
(33)		10%	WB	36.7	38.7	43.6	52.1	60.5	65.3	67.5	66.2	60.3	54.2	47.0	39.1	(33)
(34)		MADB	38.2	40.6	47.5	59.2	68.4	72.5	75.8	74.4	66.2	58.1	49.0	40.4	(34)	
(35)	Mean Daily Temperature Range	MDBR	7.8	9.5	12.4	16.3	17.9	17.7	18.1	18.6	15.9	13.2	8.8	7.3	(35)	
(36)		5% DB	MCDBR	8.0	12.5	19.0	22.5	23.0	22.7	23.8	24.4	22.0	18.7	13.6	8.7	(36)
(37)		MCWBR	7.1	9.2	12.2	12.0	12.2	11.9	10.3	10.7	11.0	11.7	9.9	7.3	(37)	
(38)		5% WB	MCDBR	7.9	11.7	17.4	21.1	21.5	20.8	21.1	22.0	19.7	17.4	12.2	8.6	(38)
(39)		MCWBR	7.1	8.9	11.9	11.6	12.0	11.6	10.5	10.2	10.6	11.6	9.4	7.4	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.312	0.339	0.375	0.429	0.416	0.414	0.439	0.434	0.398	0.364	0.347	0.310	(40)	
(41)		taud	2.242	2.226	2.216	2.127	2.224	2.257	2.201	2.214	2.276	2.347	2.302	2.291	(41)	
(42)		Ebn at Noon	217	244	257	256	266	267	258	252	247	234	202	199	(42)	
(43)		Edh at Noon	22	30	36	44	42	41	43	40	34	26	21	18	(43)	
(44)	All-Sky Solar Radiation	RadAvg	248	485	873	1305	1639	1755	1656	1483	1028	602	263	177	(44)	
(45)		RadStd	42	72	116	153	169	140	171	112	157	95	29	30	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.90	N/A	N/A	+1.55	N/A	N/A	N/A	-299	+144	+58
(47)	Regional (20 neighbors)	+0.99	N/A	N/A	+1.44	+0.74	N/A	N/A	-317	+158	+65

Nomenclature: See separate page